Pontificia Universidad Católica de Chile
IA Lab

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DL Models for Audio Processing

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DL Models for Audio

So far, we have discussed:

1. The nature of sound. In particular,
   - Its composition as a superposition of frequency components
   - Popular hand-crafted features used to process audio. Ex. spectrograms and log-mel.
2. Tools to support audio applications:
   - Structure of audio files
   - Software libraries to process audio
   - Datasets.

This class: DL Models to process audio

Applications

Audio Main Applications

According to the nature of audio signals, we can divide their applications into 3 main categories:

- General sounds.
  Ex. environmental sound classification, audio tagging, ...
- Speech.
  Ex. Speech recognition, speech translation, speaker identification, ...
- Music.
  Ex. song recognition, musical instrument identification, ...

This class mostly focuses on environmental sounds. Next class we will discuss speech and voice. We will discuss music just briefly.

General Sounds

Audio Applications: General Sounds

Sound enhancement:

- Noise reduction/elimination.
- Sound reconstruction.
- Source separation.
- Source transformation.
- ...

Audio Applications: General Sounds

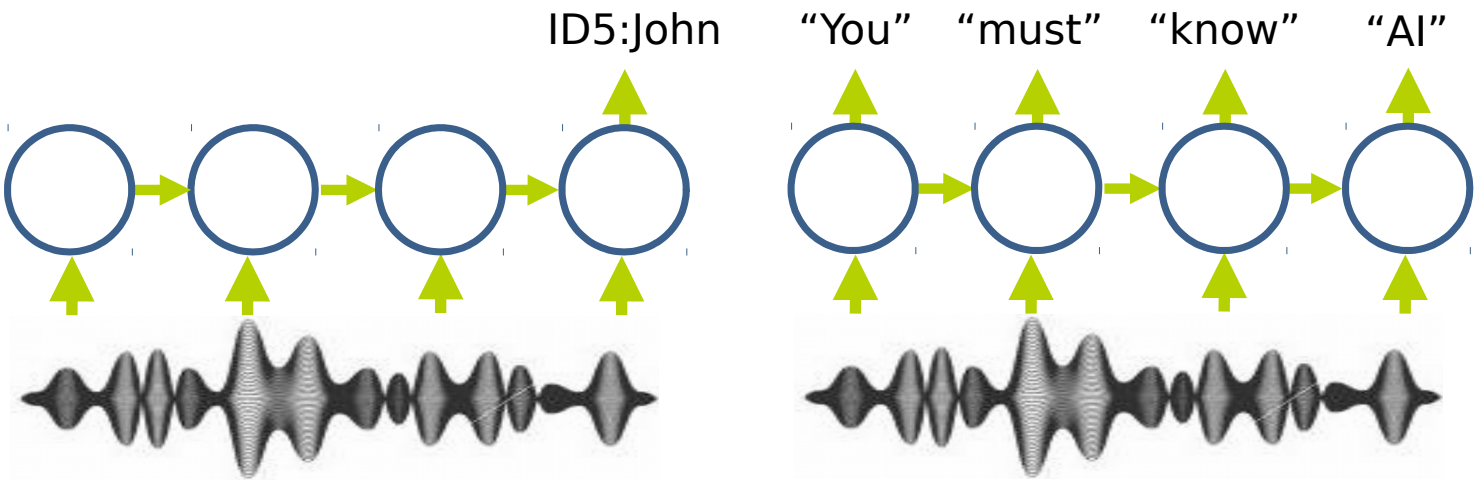
Synthesis:

- Sound synthesis.
- Speech synthesis.
- Music creation.
- ...

Audio Applications: General Sounds

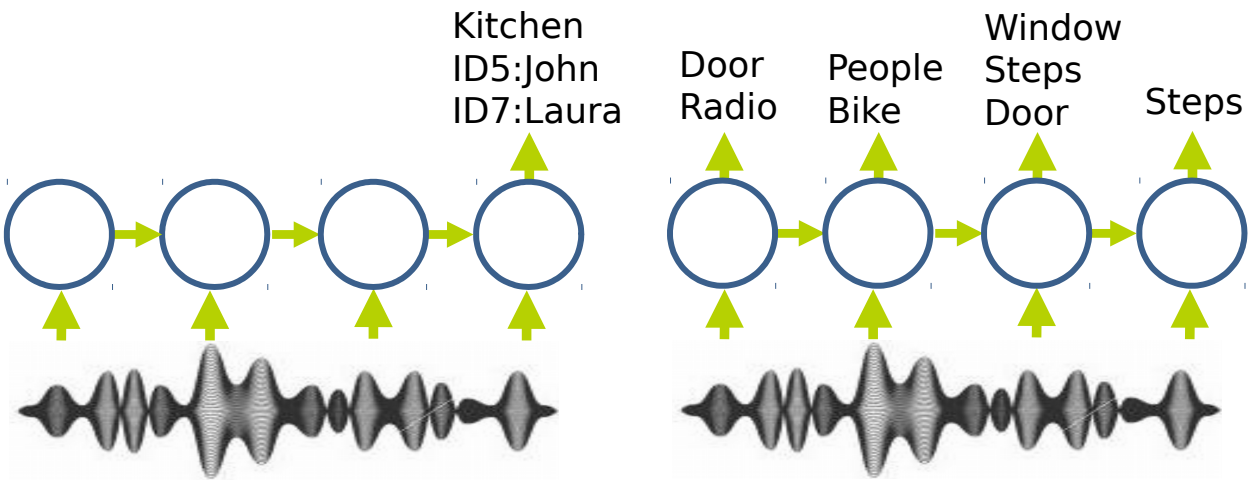
Classification:

- Single global-label vs single local-labels:
  - Ex.: John is talking vs John say: "you must know AI".



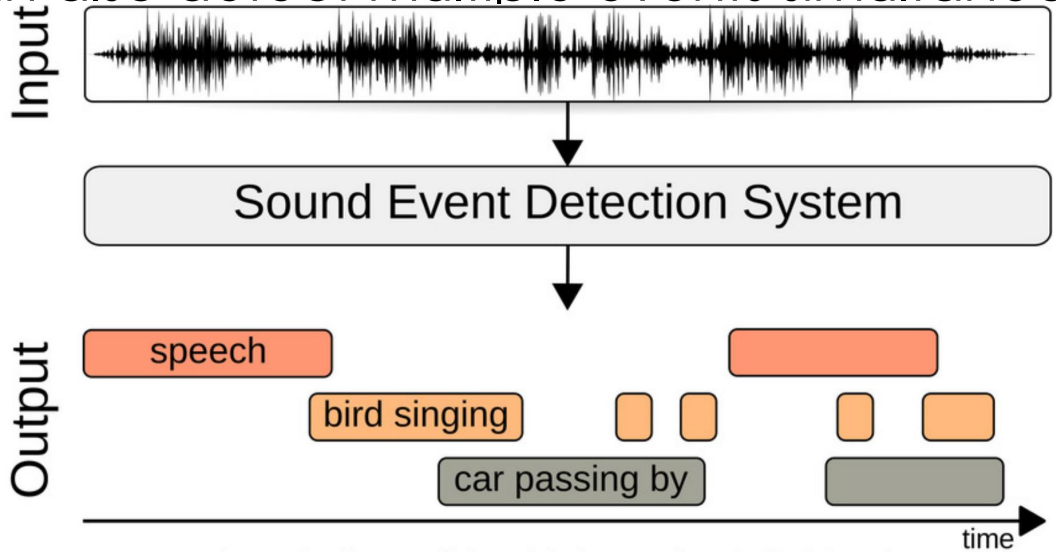
Audio Applications: General Sounds

- Multiple global-labels vs multiple local-labels:
  - Temporal tagging of audio sequences.
  - Multiple speaker identification.
  - Sound recognition at different levels of granularity. Ex. Car vs specific part of a car.



Audio Applications: General Sounds

- Event detection: detect a specific sound in a long audio sequence.
  - Somebody is opening a door.
  - There is a heartbeat anomaly.
  - We can also detect multiple events simultaneously.



Example: http://soundnet.csail.mit.edu/

Speech

Audio Applications: Speech

- Speech recognition.
- Video captioning.
- Speaker audio separation.
- Language translation.
- ...

Music

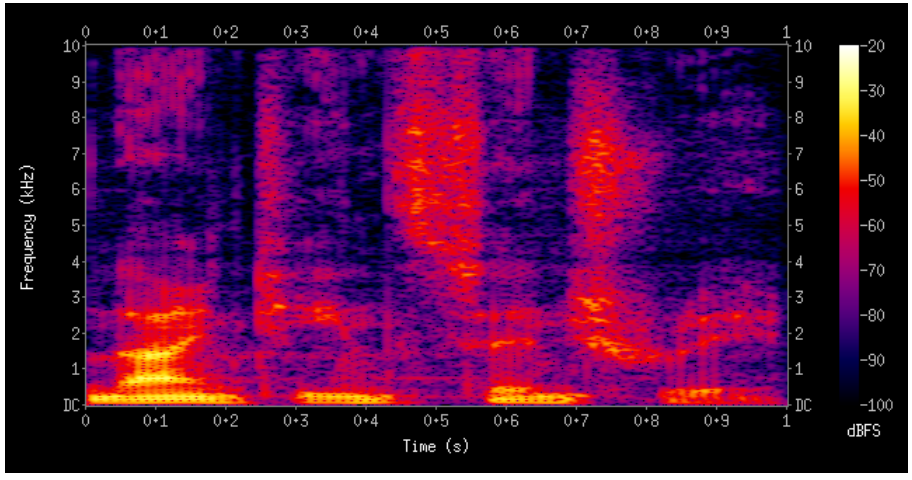
Audio Applications: Music

- Song recognition.
- Song/music-style similarity.
- Music instrument detection.
- Music transcription.
- ...

Models

Audio vs Image Data

- We have already analized models for visual recognition.
- In principle, the 2D time-freq representation (spectrogram) of an audio signal can be interpreted as an image:



- In this way, we can use 2D CNNs to process audio signals.
- While this is possible, there are relevant differences between audio and visual data that is important to consider.

Modeling Options for Audio Signals

So far, we have discussed several modeling options, which ones?

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- Transformer and relational models
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- Imitation learning
- Reasoning: Neuro-Symbolic models

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Why?

They have complementary properties to analyze audio signals.

Let's check more on this.

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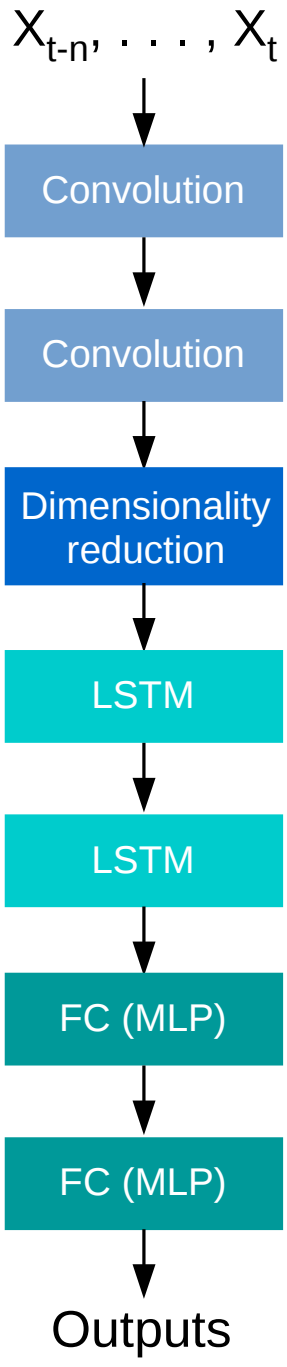

- Details of the structure are usually chosen experimentally based on validation error.
- Sample frequency and application determine window for temporal context.
- Use fewer parameters for less data to avoid overfitting.
- Decrease filter size and increase number of channels for deeper layers.

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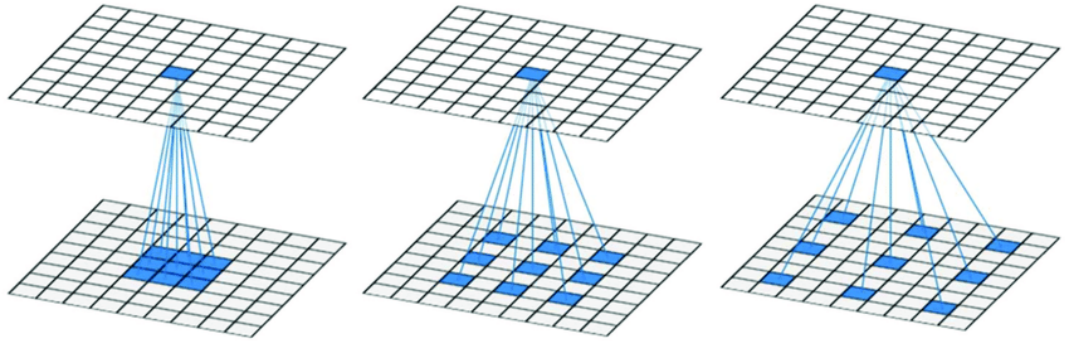
  - Dilated convolution filter ?

Dilated Convolution



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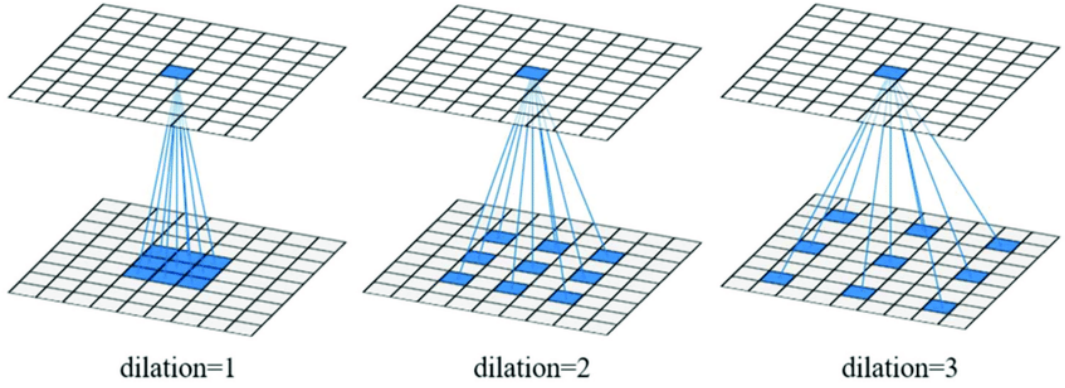

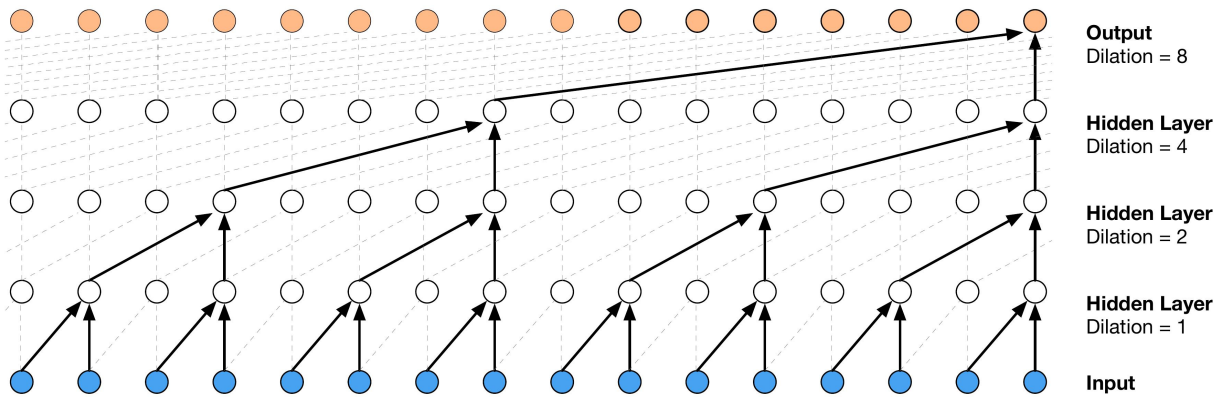

- For a CNN, reaching a sufficient receptive field size leads to a large number of parameters and a high computational complexity.
- Dilated convolutions enable CNNs to have very large receptive fields with just a few layers depth.
- In case of raw audio data, after few layers depth, neurons can cover thousands of timesteps, keeping a suitable computational efficiency.

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- 4 dilated convolutional layers (dilation factor depends of application). Each with: i) 128, 128, 256, 256 filters, ii) 20x1, 10x1, 10x1, 5x1 filter sizes, respectively.
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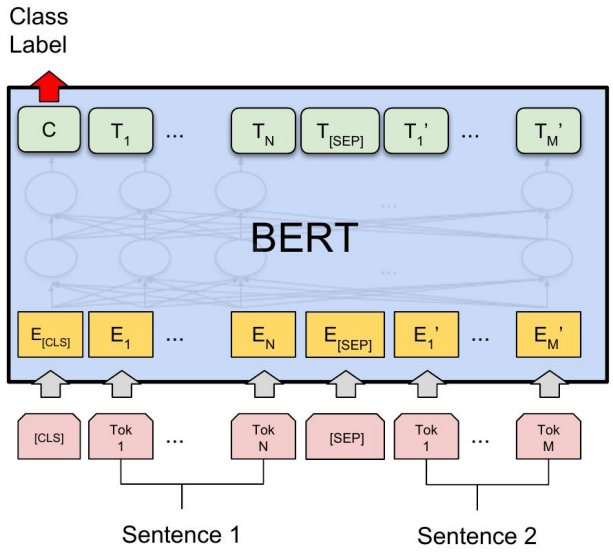
Audio and Transformers



- There have been previous works on audio applications using a Transformer architecture. However, there are 3 relevant problems:
  1. In the context of audio, there is still a lack of highly massive audio datasets.
  2. Self-attention mechanism operates over a finite sequence of discrete entities. In the context of text, sentence segmentation is trivial, but for audio this is not the case.
  3. Transformers are not good to model long dependencies in sequences.
- As a consequence, Transformers are not currently very popular for audio applications.

Audio: Data Augmentation

Data augmentation generates additional training data by manipulating existing examples.

- In the case of audio, typical strategies include:
  - Modify pitch.
  - Add noise.
  - Perform time stretching.
- In speech recognition, a model can be pretrained on languages with more transcribed data and then adapted to a low-resource language or domain.
- Use of sound synthesis techniques is a convenient strategy. However, performance on real data might be poor if a model is trained using only synthetic data. Need for finetunning on real data.

References

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